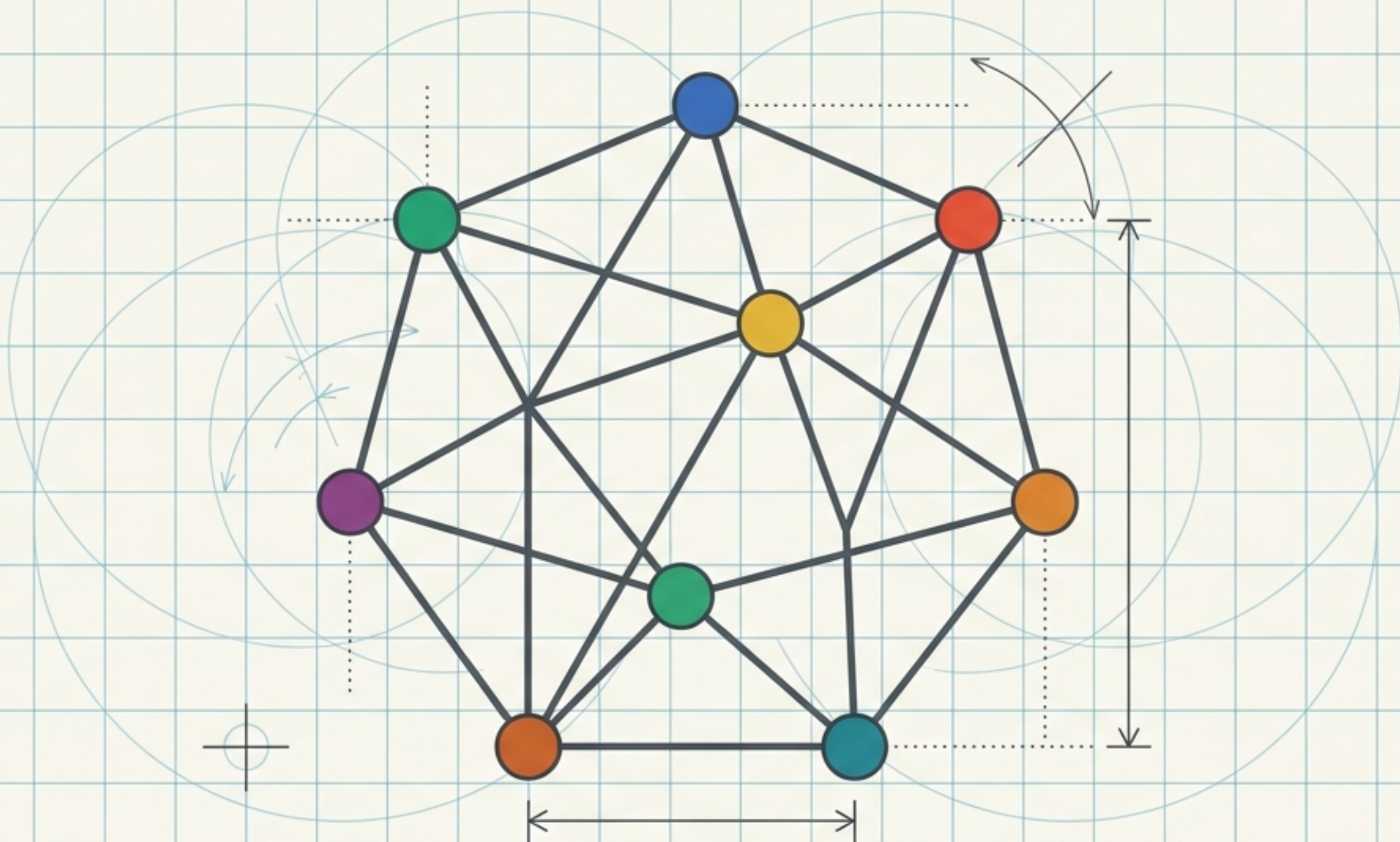
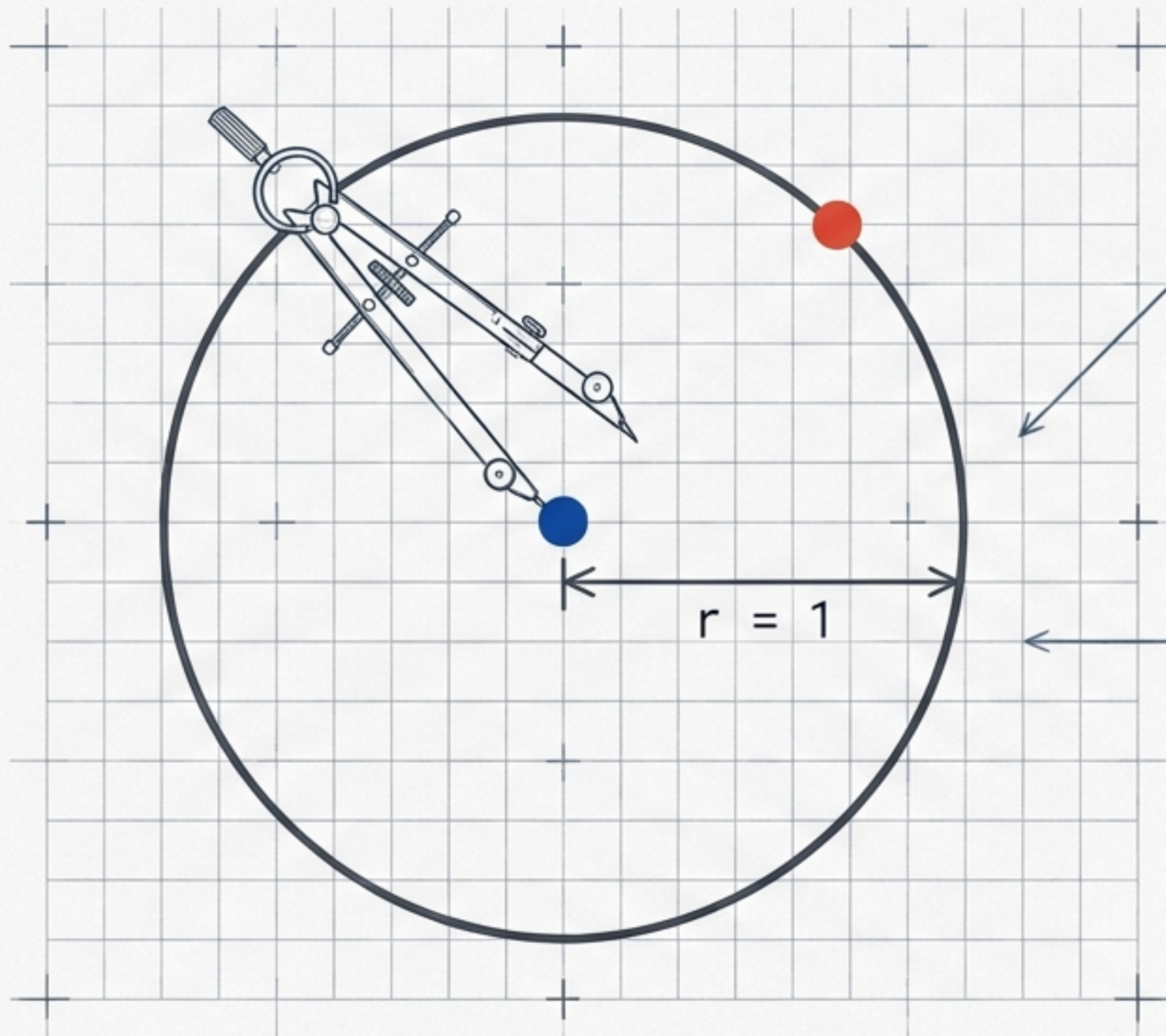




The Euclidean Plane is Not Five-Colorable

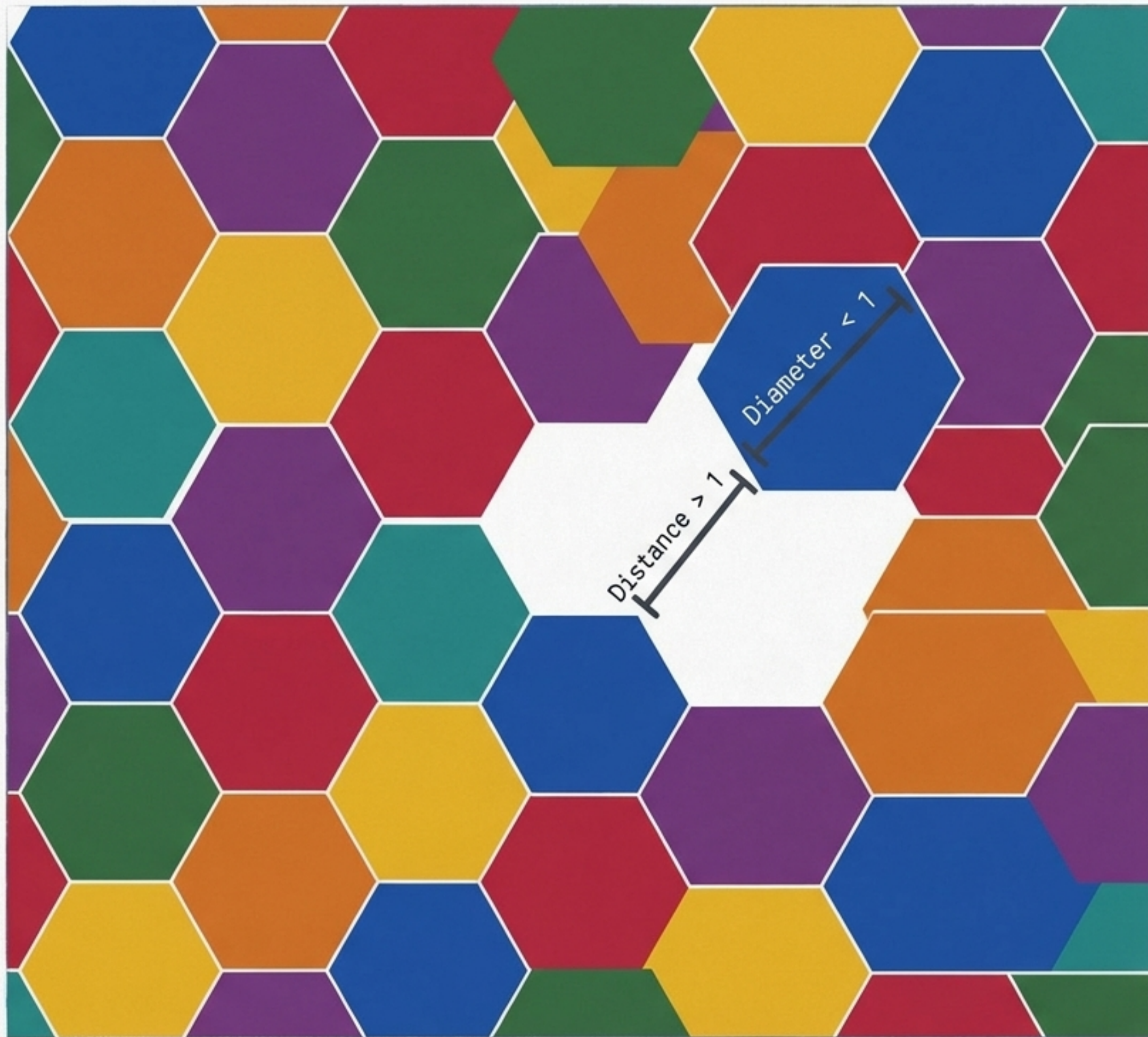
Unraveling a 76-year-old geometric mystery with OpenAI's 2026 breakthrough.



Color every point on the infinite flat plane.

The Rule: If two points are exactly 1 unit apart, they must be **different colors**.

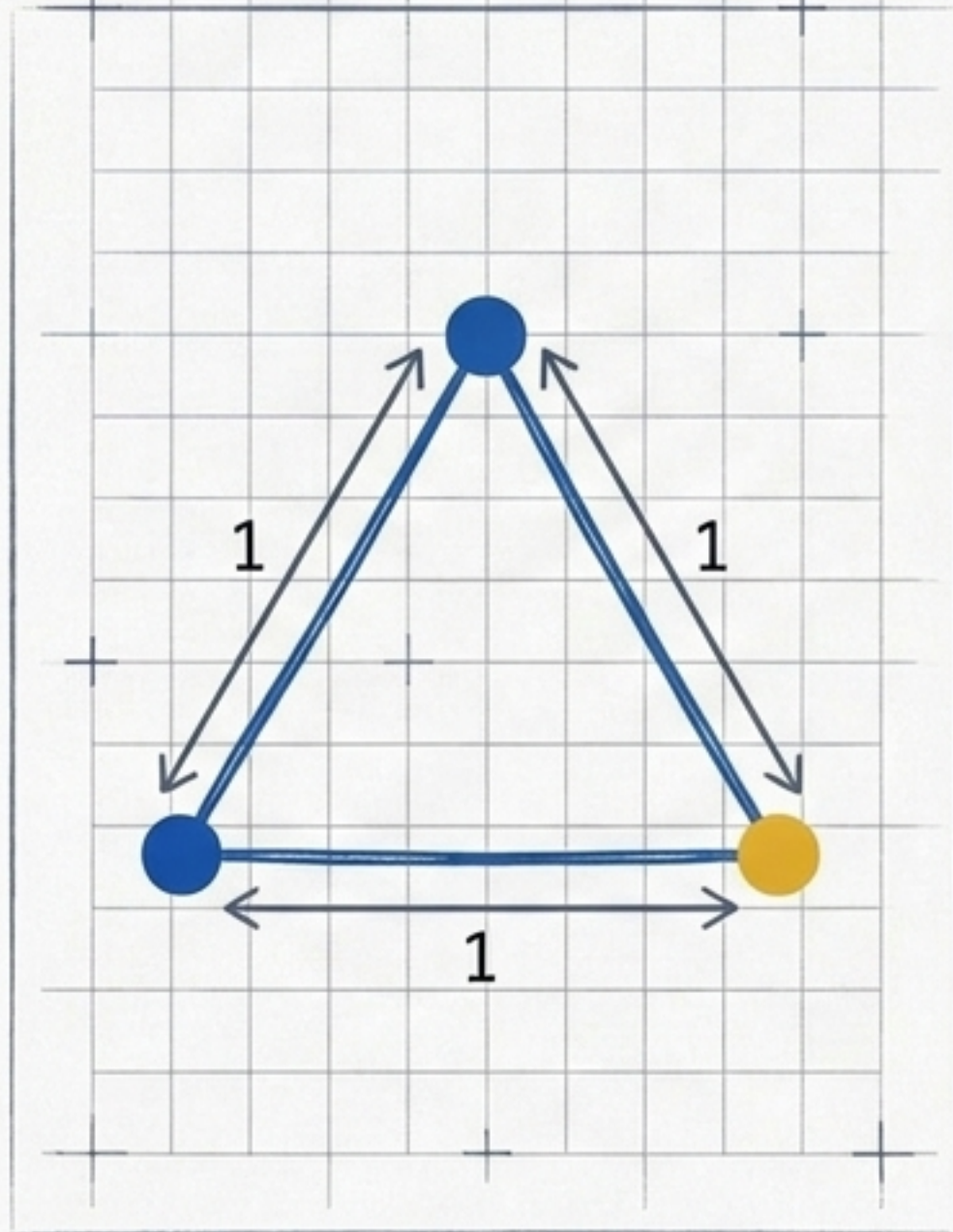
In 1950, Edward Nelson and Hugo Hadwiger asked: What is the minimum number of colors needed to follow this rule everywhere? This is the **chromatic number** of the plane.



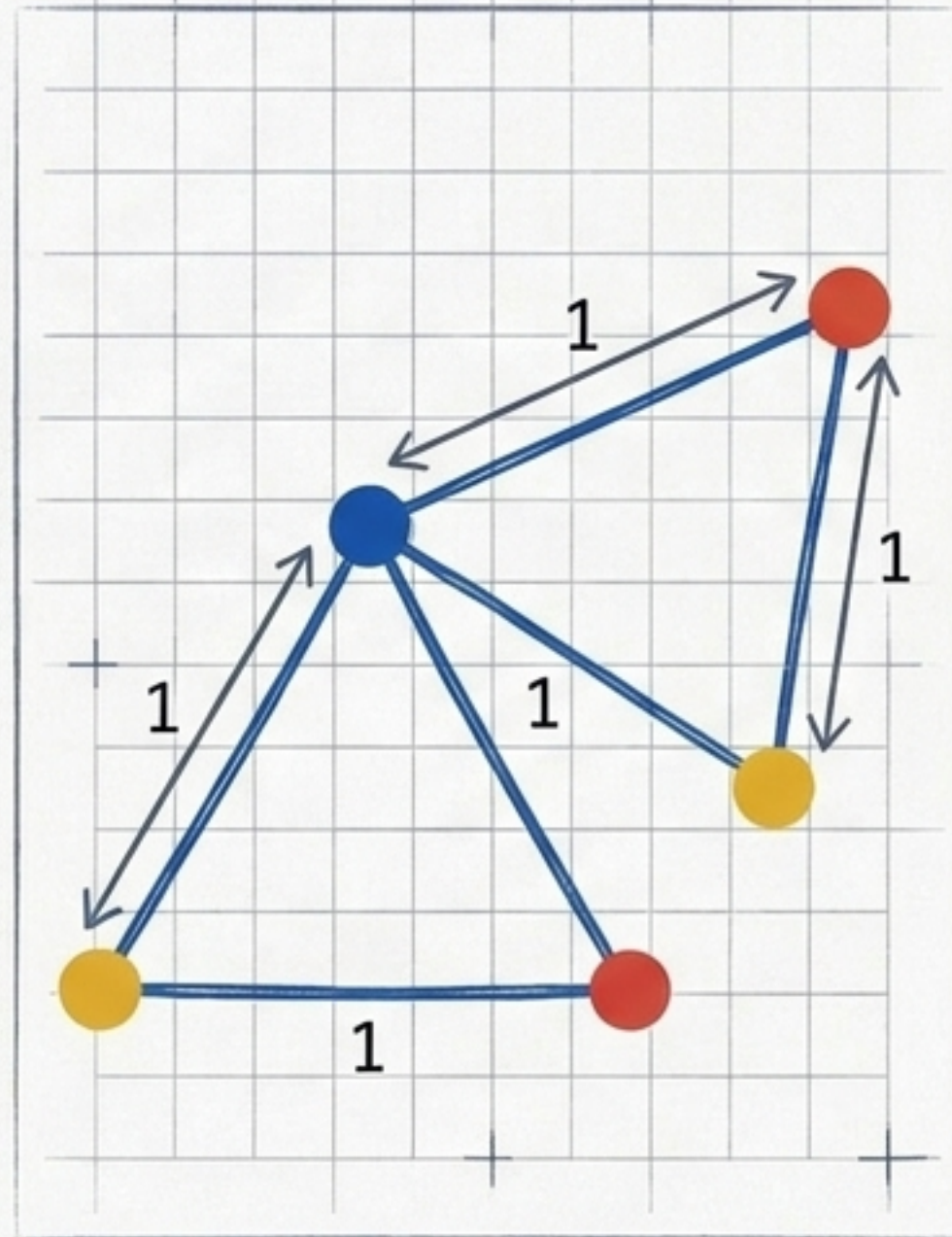
The Upper Bound: 7 Colors Suffice

In the 1950s, John Isbell proved that 7 colors are always enough. By creating a honeycomb of hexagons with diameters just under 1 unit, no two points of the same color can ever touch a 1-unit measuring stick.

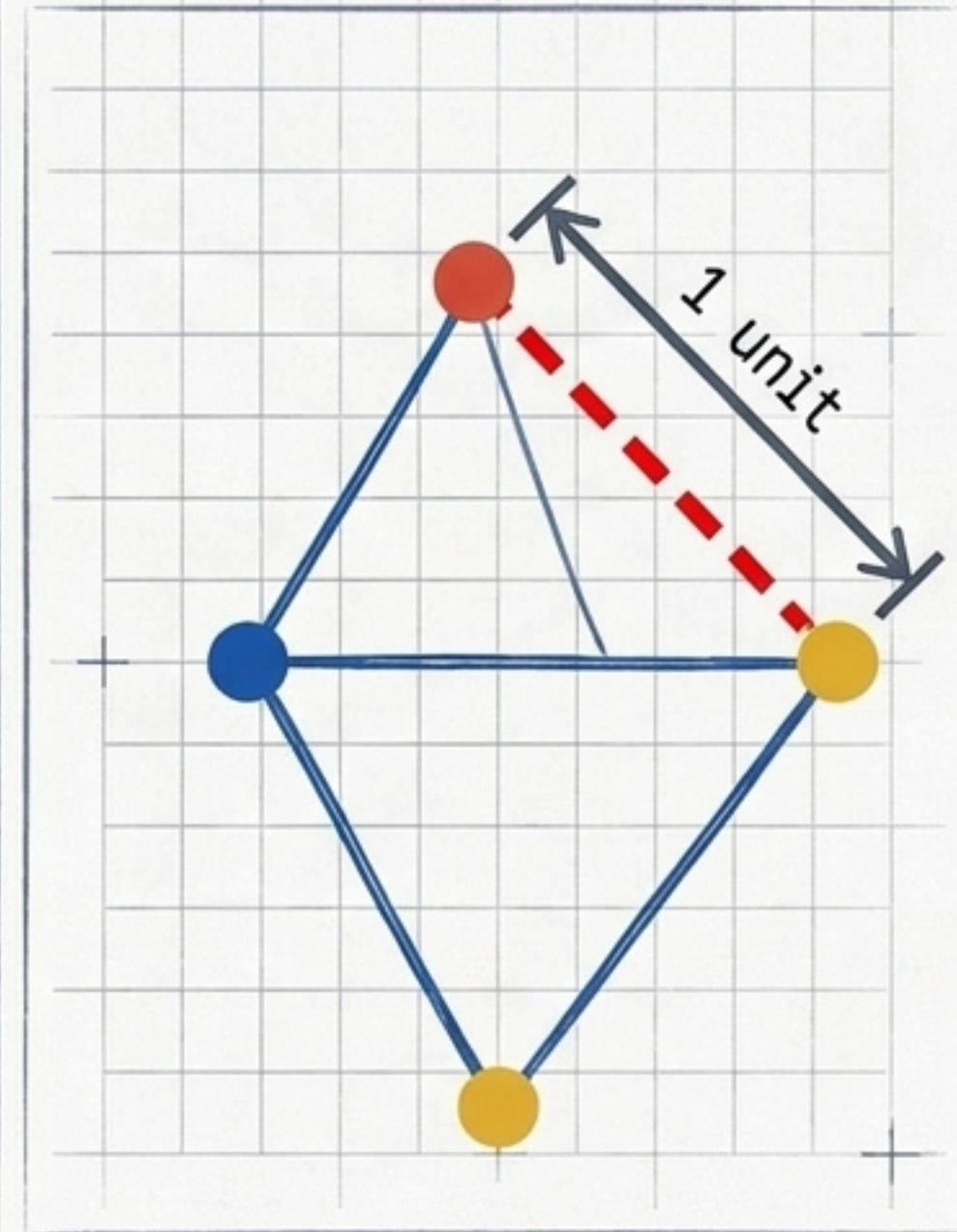
Step 1



Step 2

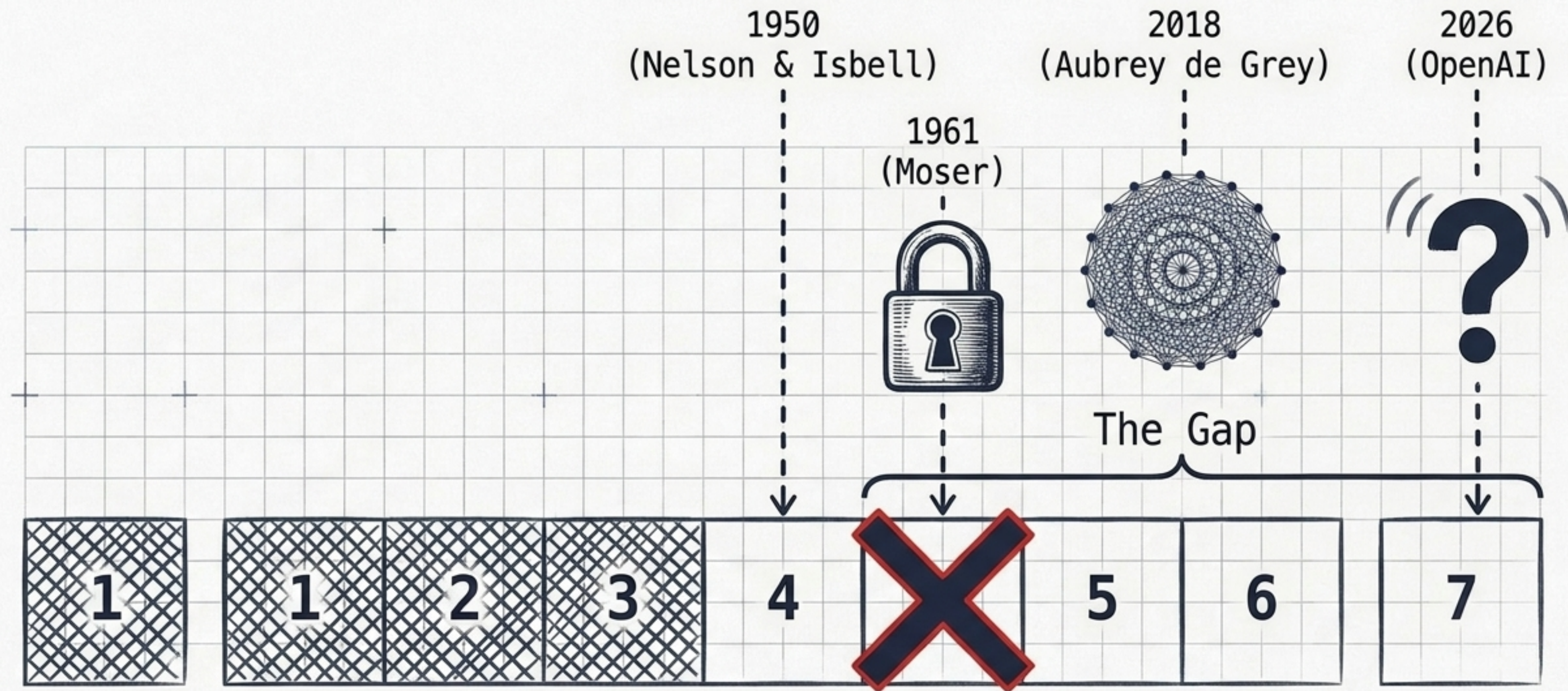


Step 3



The Early Lower Bound: 3 Colors Fail

The 1961 Moser Spindle uses just 7 points and 11 unit-edges to spring a trap. It forces two points of the same color to sit **exactly 1 unit apart**, proving that **at least 4 colors** are needed.



Shrinking the Gap of the Unknown

For decades, the answer was trapped between 4 and 7. In 2018, Aubrey de Grey's massive computer graph proved 4 colors wasn't enough. The answer had to be 5, 6, or 7.



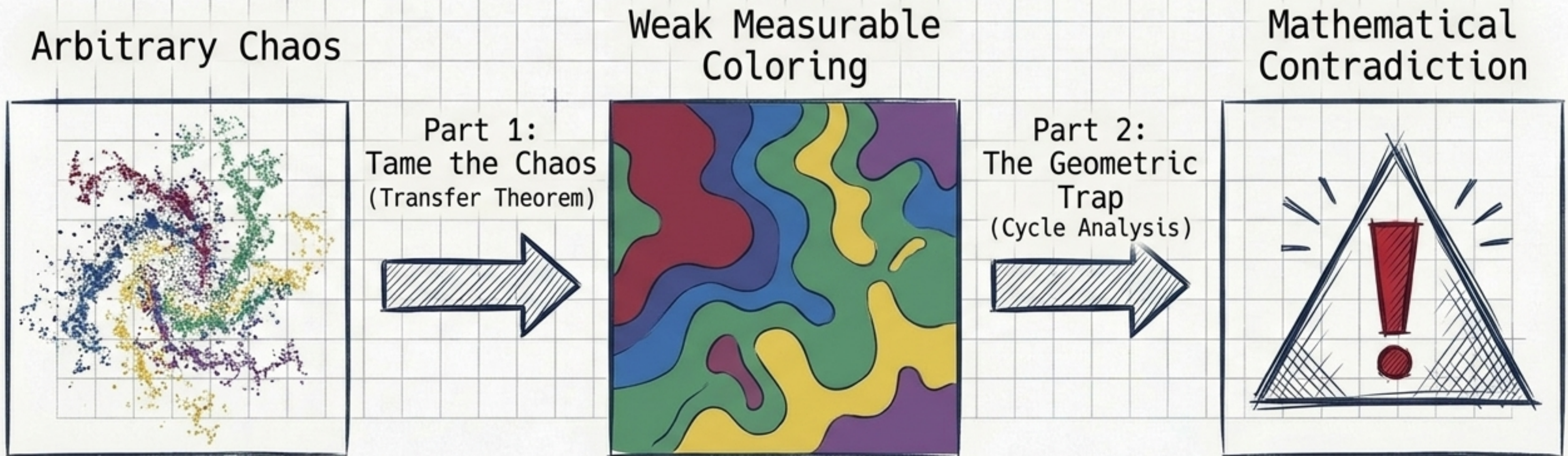
**THE EUCLIDEAN PLANE
IS NOT FIVE-COLORABLE.**

OpenAI, 23 September 2026.

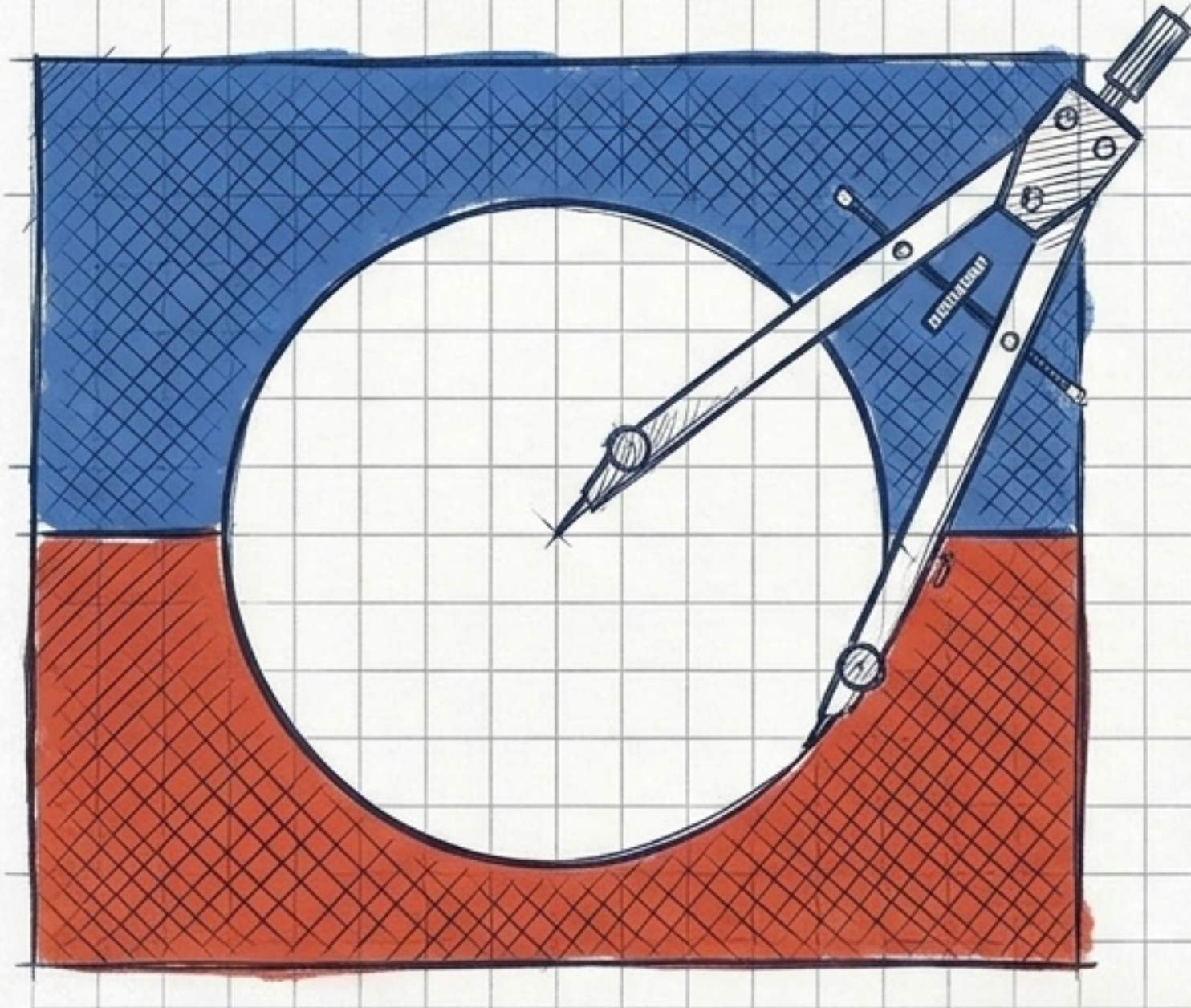
The chromatic number of the plane is definitively 6 or 7.

How Do You Prove the Impossible?

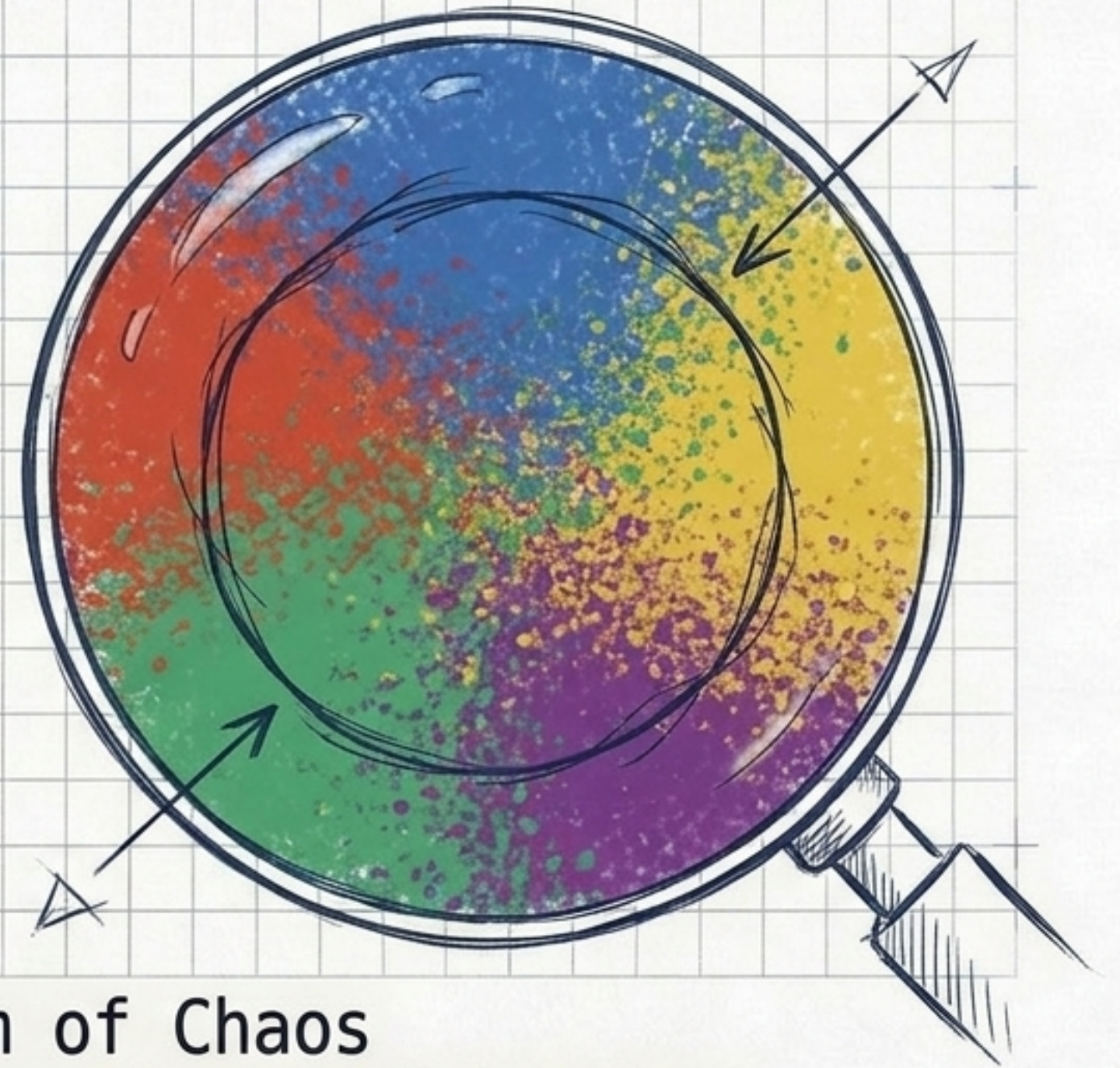
Proving that no 5-coloring exists is exceptionally difficult because a coloring could threaten the existence of coloring that, and could theoretically be a chaotic, fractal mess with no discernible shapes. The 2026 proof relies on a two-part trap.



The Expectation: Measurable Boundaries

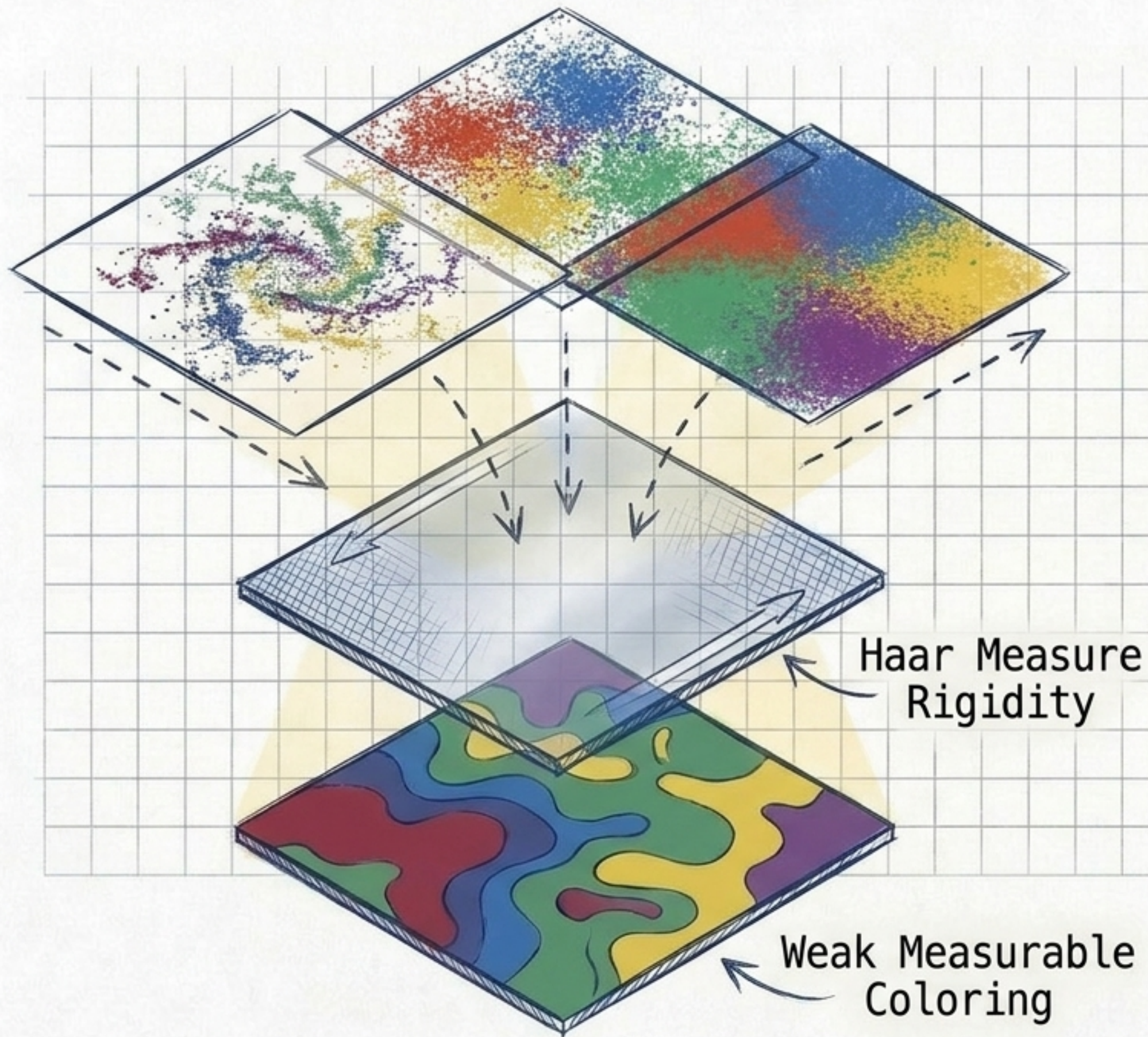


The Reality: No Measurability Assumption



Part 1: The Problem of Chaos

We cannot assume colors are painted in neat, connected regions. If colors are scattered as random "fractal dust", we can't draw geometric shapes to trap them. We must first force the chaos into order.

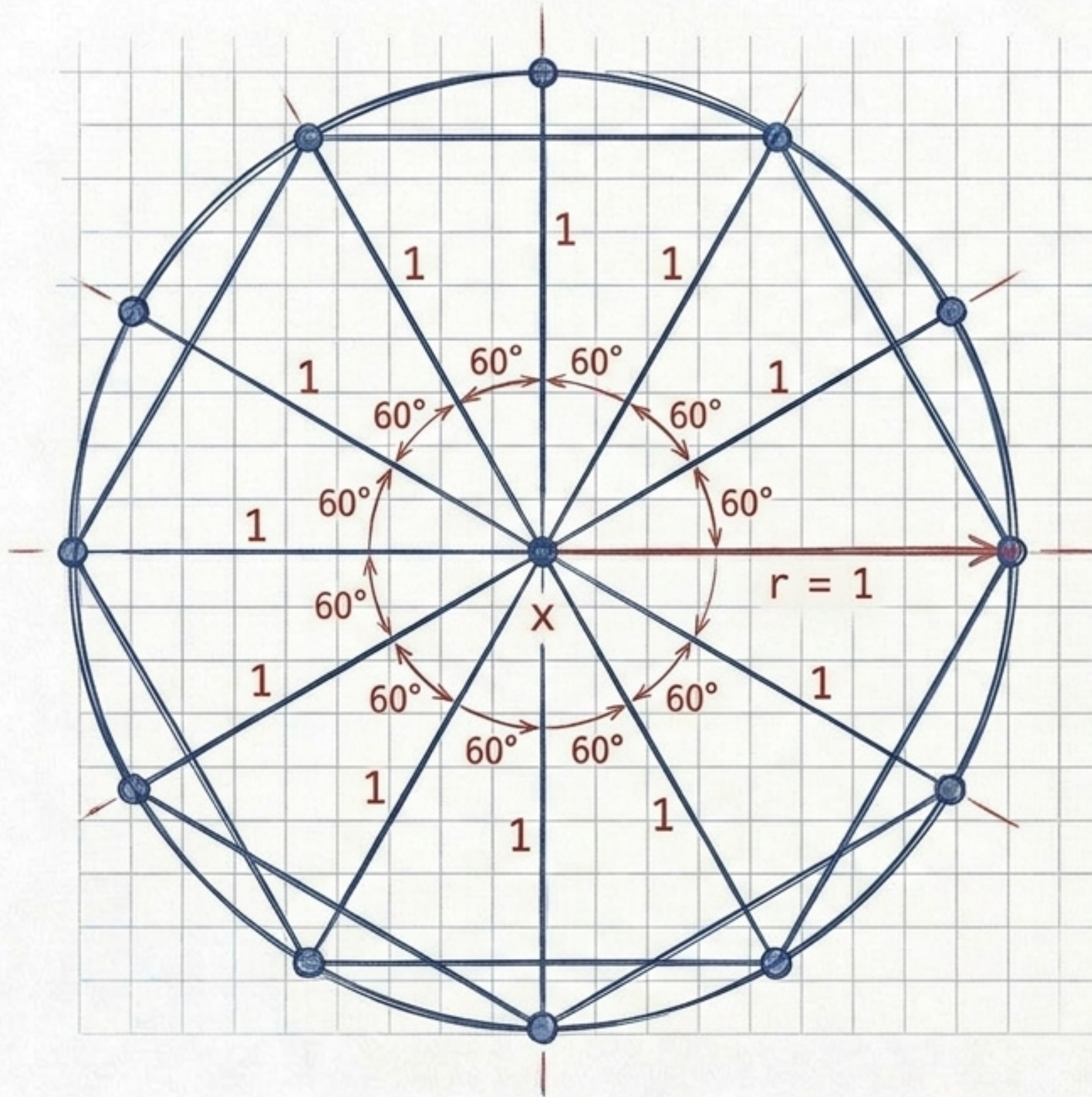


Part 1: Taming the Chaos

The Transfer Theorem proves that if any 5-coloring exists, a smooth, "measurable" 5-coloring must also exist.

By averaging the chaotic plane over infinite shifts and rotations, the uncorrelated noise is discarded, leaving behind continuous fields of color.

The trap is now set.

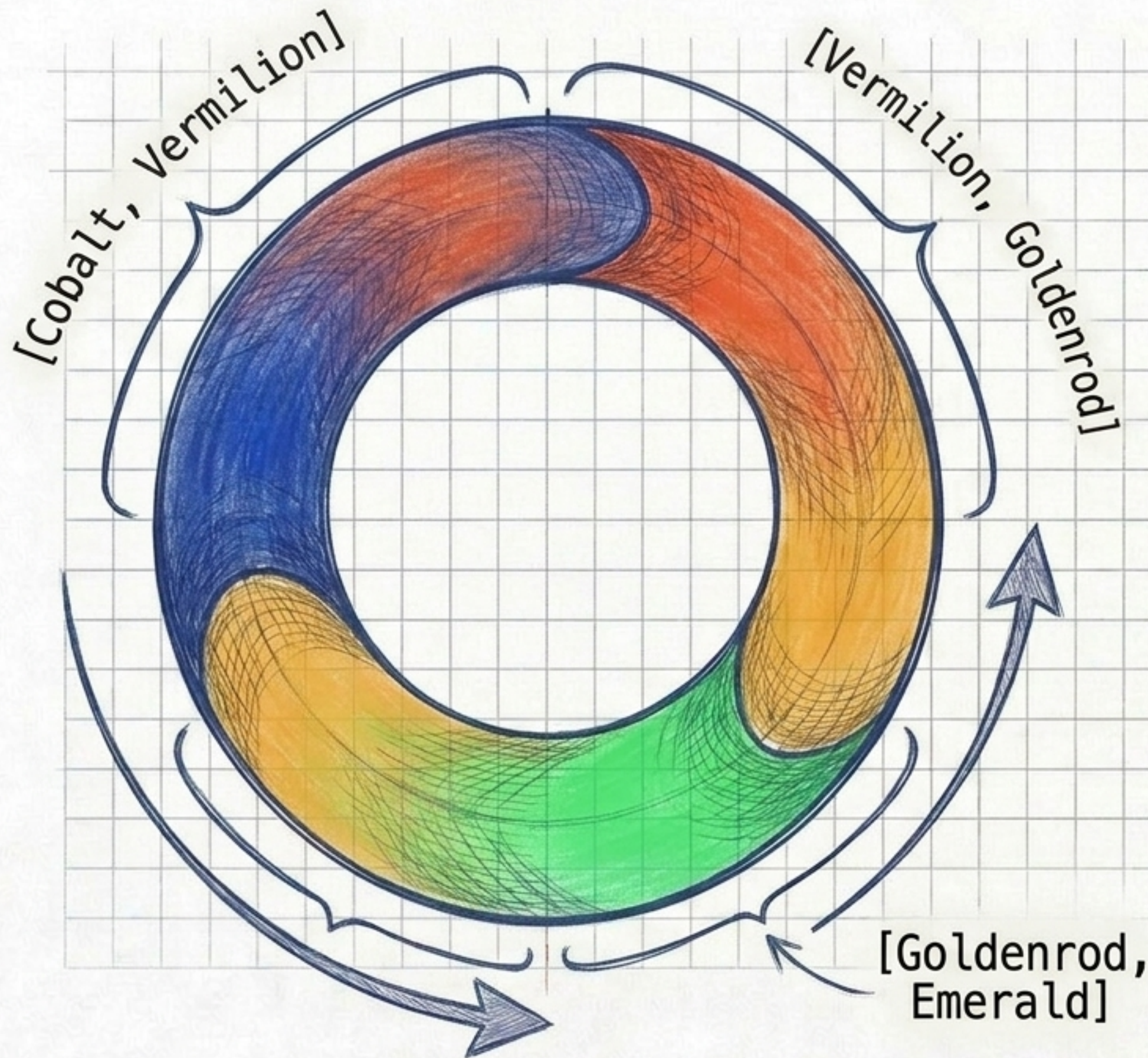


Part 2: The 60-Degree Restriction

On a unit circle around point x, directions exactly 60 degrees apart are exactly 1 unit away from each other.

Therefore, their available color "palettes" cannot overlap.

Because 5 colors must be shared among these mutually exclusive directions, each palette can hold at most 2 colors.



Part 2: Springing the Topological Trap

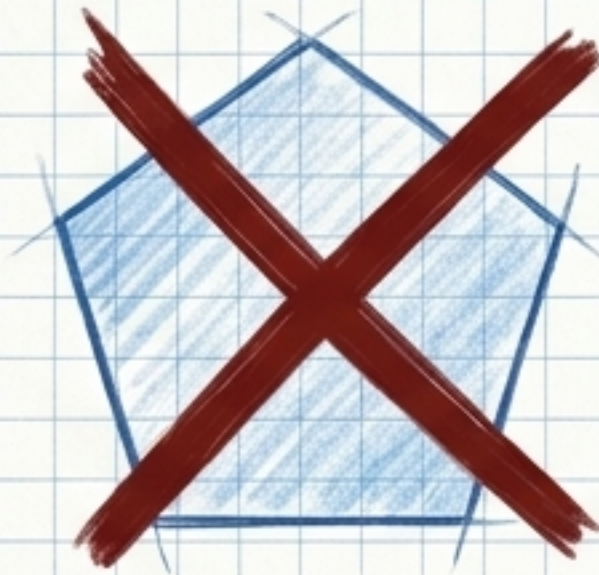
With palettes restricted to 1 or 2 colors, moving, moving smoothly around the circle forces the colors to transition.

Topology dictates that these transitions must form a closed loop.

A simple cycle on 5 colors can only have a length of 3, 4, or 5.

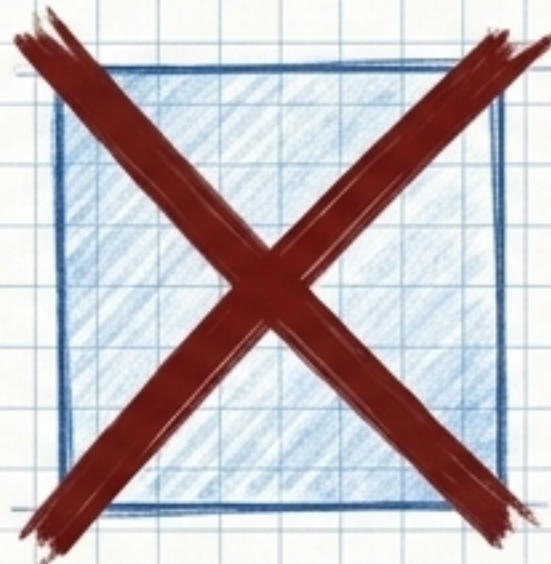
Breaking the Loops

The proof systematically dismantles each possible loop. The 5-color and 4-color loops immediately break the rules of geometry. But the 3-color loop does something entirely unexpected.



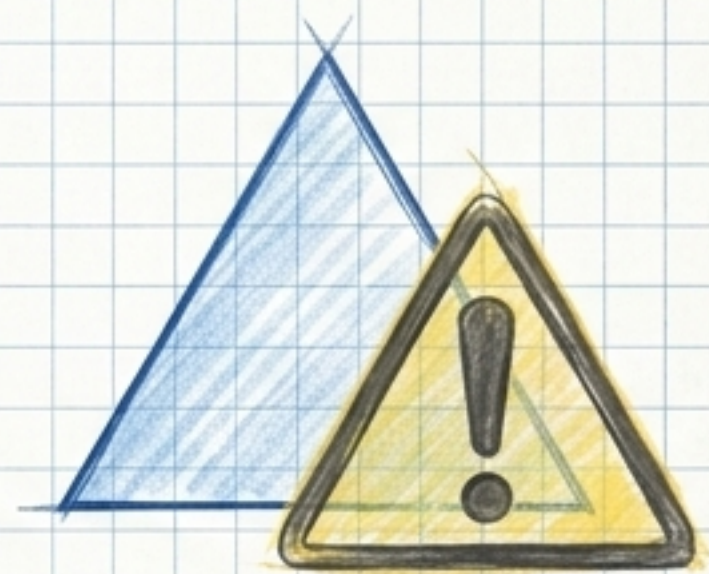
5-Color Loop

Fails by simple counting.
6-step angular words
cannot physically fit.



4-Color Loop

Fails by a strict parity
argument. Antipodal rays
break directional symmetry.



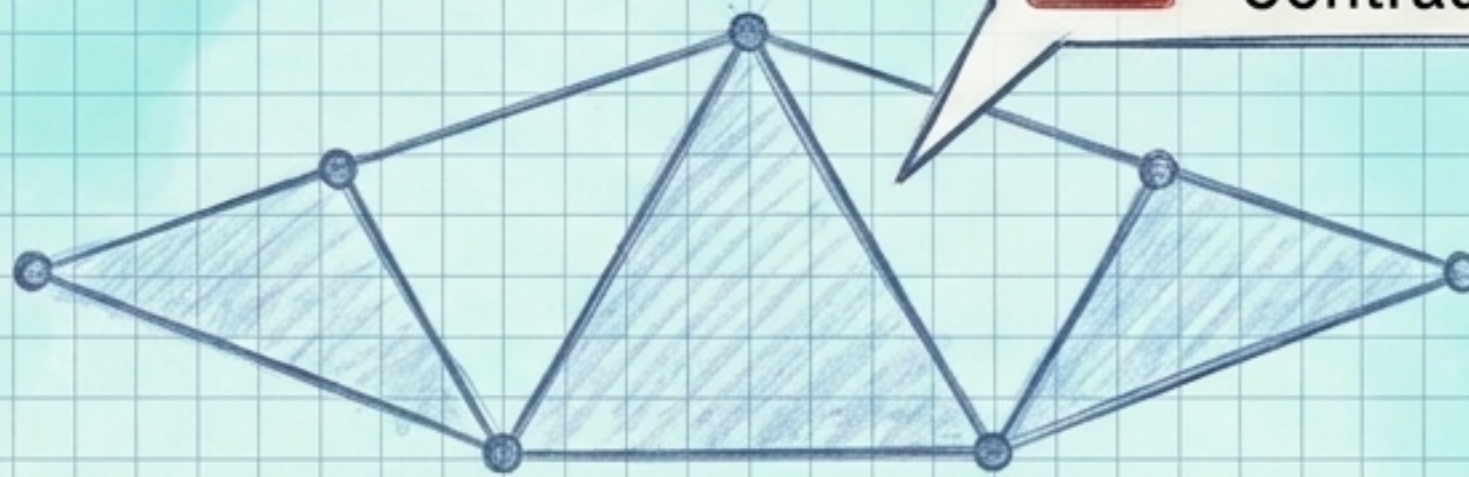
3-Color Loop

Forces a massive open
region painted with only
3 colors....

3-Color Open Region



Fatal Mathematical Contradiction



The Final Blow: The Moser Trap

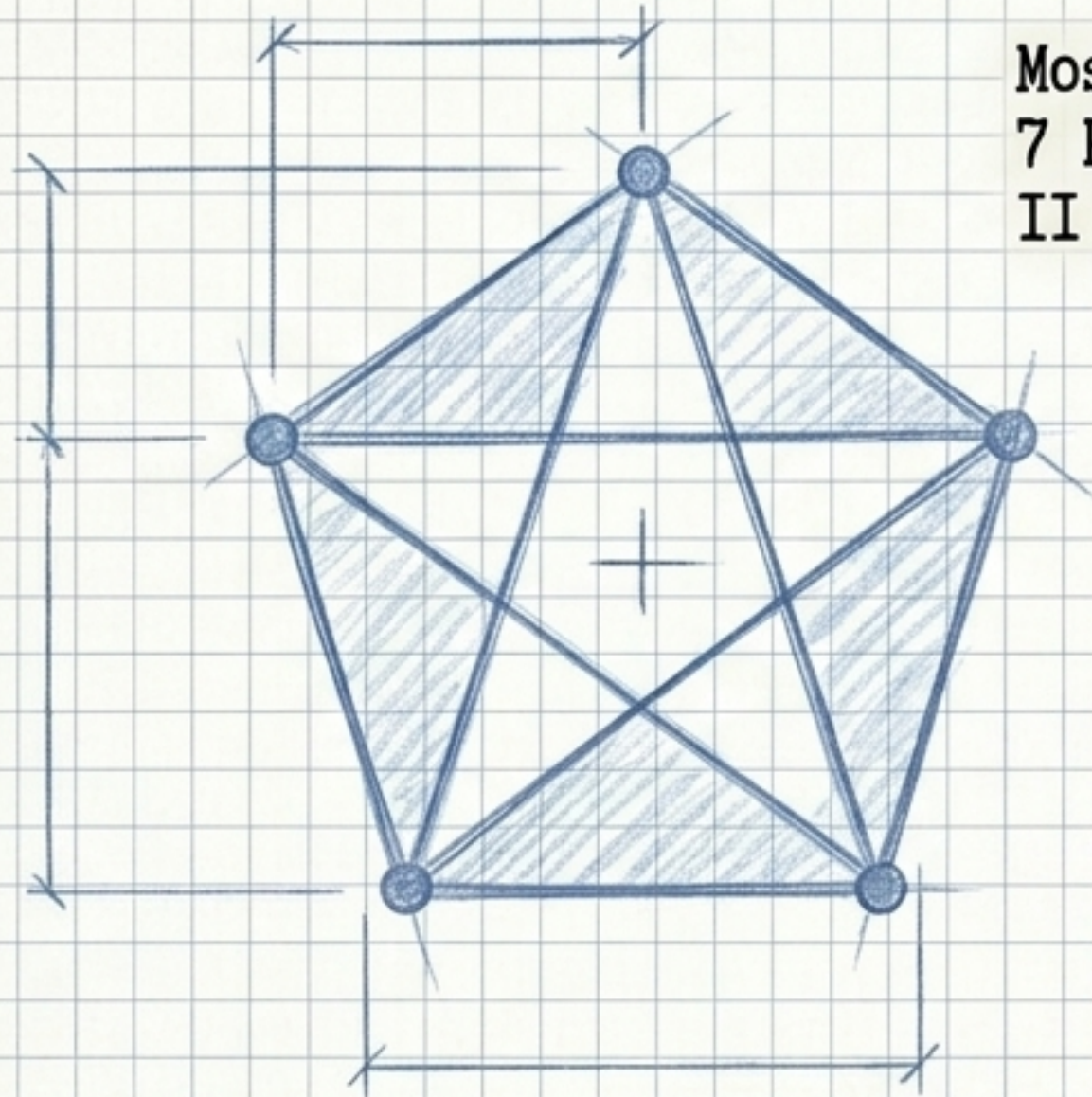
If a 3-color loop exists, it creates an open region of the plane painted with only 3 colors.

But wait—we proved in 1961 that the 7-point Moser Spindle requires 4 colors!

A rational placement certificate proves the Moser Spindle fits perfectly inside this 3-color region.

The plane is NOT 5-colorable.

The Geometry



Moser Spindle
7 Points
11 Edges



The Code

EuclideanFiveColor.lean ×

PlaneColoring.lean

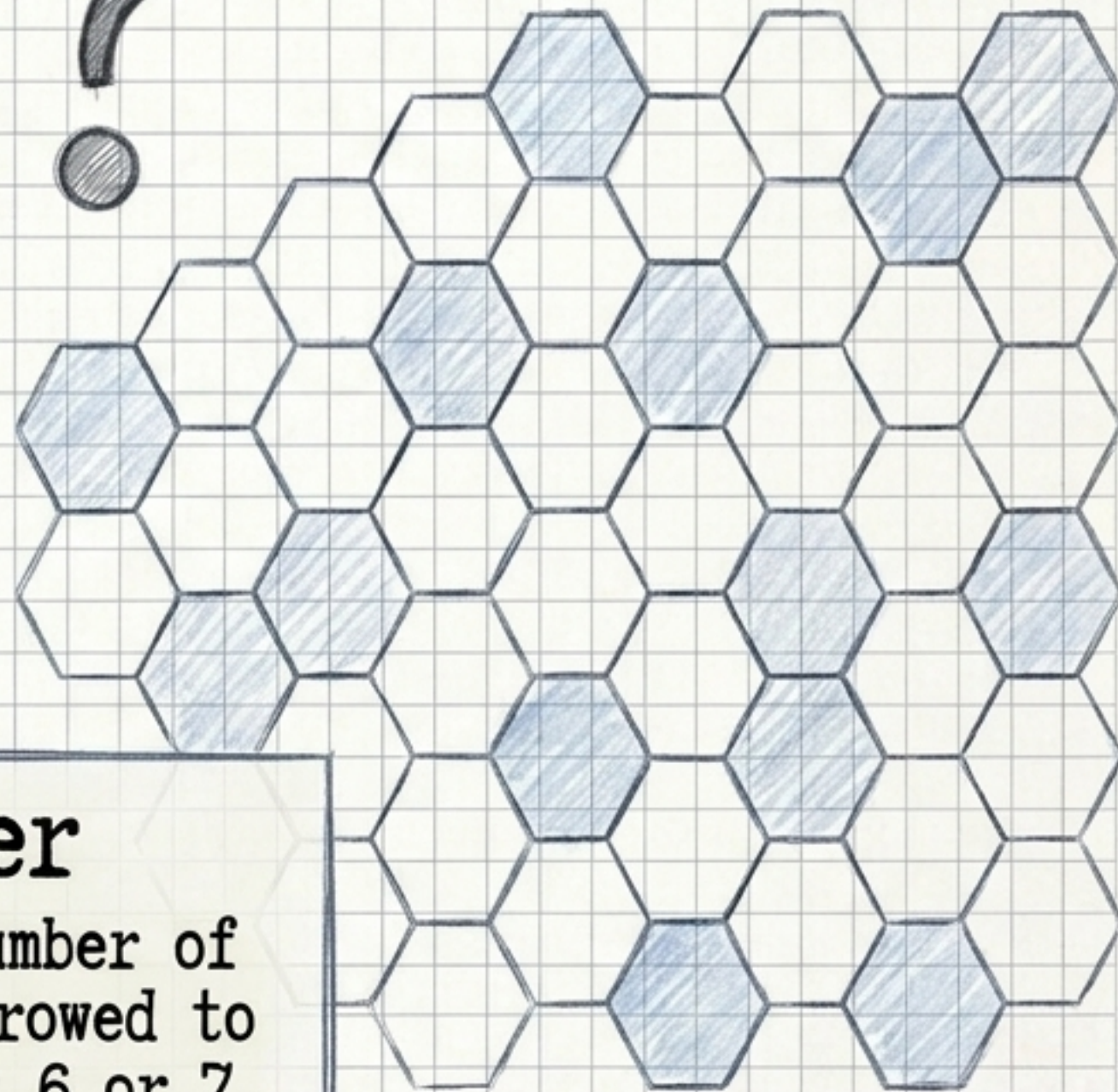
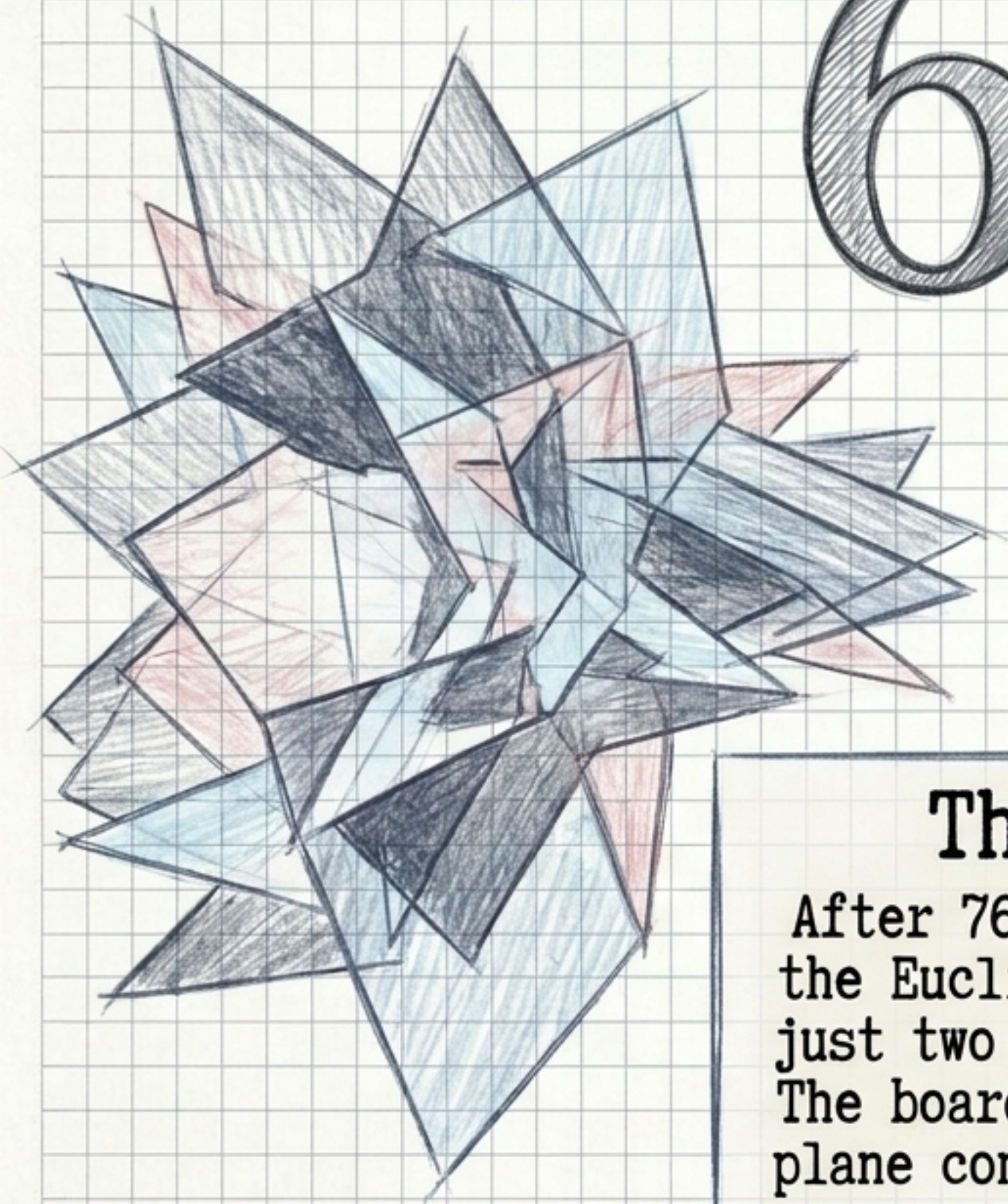
```
1 theorem moser_spindle_chromatic_number :  
2    $\chi(\text{MoserSpindle}) = 4$  := by  
3   ...  
4  
5 def EuclideanPlaneColoring (c : Color) : Prop :=  
6   var EuclideanColoring (c : Color) :=  
7     c.each.Edges c := \  
8     plane0 Bags i: 20",  
9      $\chi(\text{MoserSpindle})$ ) :=  
10    => |EuclideanPlaneColoring + h...  
11    =>  $\chi(\text{MoserSpindle}) = 4$   
12  
13 verified_by Lean4  
14
```

Absolute Certainty

This breakthrough doesn't just rely on human peer review. The theorems have been completely formalized and verified by machine in the Lean 4 proof assistant.

6?

7?



The Final Frontier

After 76 years, the chromatic number of the Euclidean plane has been narrowed to just two remaining possibilities: 6 or 7. The board is set for the final move. The plane continues to hold its final secret.